L1.4 The Derivative

§2.1–2.2 — what you need to know

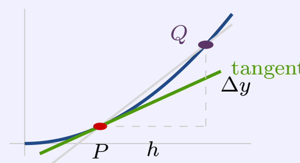
The derivative

$$\boxed{f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}}$$

"take the difference quotient, and let h go to zero"

The fraction on its own is the **difference quotient** — the slope of the **secant** through P and Q. Shrink h and Q slides toward P.

At $h = 0$ there is no second point and no secant at all. That is why it has to be a *limit* and not an evaluation.



$$f'(x) = \frac{dy}{dx} = \frac{d}{dx}\big(f(x)\big) \qquad \text{evaluate with } \left.\frac{dy}{dx}\right|_{x=3} \qquad \text{second derivative } f''(x) = \frac{d^2y}{dx^2}$$

dx and dy are called **differentials** — $\frac{dy}{dx}$ is not one symbol. The book also writes $\lim\limits_{x\to a} \dfrac{f(x) - f(a)}{x - a}$ and $\lim\limits_{\Delta x\to 0} \dfrac{\Delta y}{\Delta x}$; same derivative, and we compute with the boxed one.

EQUATION OF THE TANGENT LINE

The derivative is a slope. If the question asks for the *line*, you have one step left. (`TL` is the shorthand.)

For $f(x) = 2x^2 + 1$ at $x = 2$: $f'(2) = 8$ and $f(2) = 9$, so

$$y = mx + b: \qquad 9 = 8(2) + b \;\Rightarrow\; b = -7 \qquad \Longrightarrow \qquad \boxed{y = 8x - 7}$$

Three different objects, handed back for each other: **secant slope** · **tangent slope** · **the equation of the tangent line**.

THE GRAPH OF f' FROM THE GRAPH OF f

f' is a graph of the slopes. That one sentence generates the whole table.

`HT` on f → a **zero** on f'	f increasing → $f' > 0$
`IP` on f → an `HT` on f'	f decreasing → $f' < 0$

f'' is the **concavity** — how strongly the curve is bent. Arcing downward gives $f'' < 0$, **cup down**; arcing upward gives $f'' > 0$, **cup up**. *Make the cup with your hands.* Where the concavity switches is an **inflection point**.

POSITION, VELOCITY, ACCELERATION

Position $x(t)$ — where you are, measured from a chosen origin. **Velocity** is its derivative, and **acceleration** is the derivative of that.

$$v(t) = x'(t) = \frac{dx}{dt} \qquad\qquad a(t) = v'(t) = x''(t) = \frac{d^2x}{dt^2}$$

For $x(t) = t^3 - t$: $\quad v = 3t^2 - 1 \quad$ and $\quad a = 6t$.

$\frac{\Delta x}{\Delta t}$ over an interval is **average** velocity (a secant slope); $x'(t)$ at one instant is **instantaneous** velocity (a tangent slope). Velocity is signed: negative means heading back toward the origin.

DIFFERENTIABLE AT a

$$\boxed{f \text{ is differentiable at } a \iff f'(a) \text{ exists}}$$

On an open interval, differentiable at every point of it. **Nothing here is about f' being continuous.**

The standing example. $|x^2 - 4|$ flips branches where $x^2 - 4$ is negative, i.e. $|x| < 2$ — so $f' = 2x$ outside and $-2x$ inside. Two corners, at $x = \pm 2$: differentiable on $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$

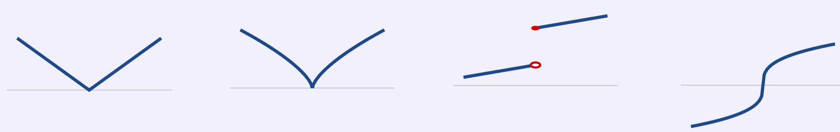
Kinds of non-differentiability

① **corner** f' **jumps** — the one-sided slopes are different numbers

② **cusp** f' has a **vertical asymptote** — sides run to $+\infty$ and $-\infty$

③ **jump** in f f' is **not there at all** — f is not even continuous

④ **vertical tangent** f' runs to $\pm\infty$ — the tangent is vertical, so it has no slope



corner cusp jump vertical tangent

WATCH OUT

✗ ~~$f'(2) = 8$ is the equation of the tangent line~~ → **it is the *slope* — the line is $y = 8x - 7$**

✗ ~~the difference quotient is the derivative~~ → **that is the *secant* slope — take the limit**

✗ ~~continuous means differentiable~~ → $|x|$ **is continuous at** 0 **and not differentiable there**

✗ ~~a corner and a cusp are the same failure~~ → **corner: f' *jumps*. cusp: $f' \to \pm\infty$**

✗ ~~$\frac{dy}{dx}$ is one symbol~~ → dx **and** dy **are differentials**

✗ ~~where f has a maximum, f' has a maximum~~ → **where f has an HT, f' has a *zero***